**Project:** Azure Capacity Reservation Management Engine (ACRME)
**Classification:** Principal Cloud Architect - Architecture Governance
**Version:** 2.4
**Date:** 7 September 2026
**Status:** Accepted - supersedes ADR-001 v2.2 region-selection content; reconciled to Baseline v2.4
**Part of:** ACRME Architecture Decision Records - aligned to Capacity & Quota Management Requirements Baseline v2.4.

> **About ADRs.** An Architecture Decision Record captures a significant architectural decision, the context that forced it, the options considered, the choice made, and its consequences. This ADR states the accepted region-selection decision matching the consolidated requirements baseline — notably the five-geography region model with per-geography distribution models and mandatory two-region CVAL/DR co-location (PLC-010a). Evidence tags: `[Documented]`, `[Decided]`, `[Derived]`, `[Assumed]`.
>
> **v2.4 reconciliation note — no region-model change.** Baseline v2.4 folds in the reservation-model gaps (seed-at-0 matrix CAP-022, per-AZ CRG structure CAP-023, reactive discovery CAP-024, Availability-Set ineligibility CAP-020/021, even zone-distribution PLC-011, naming convention OPS-006/C-12). **None of these change the region-selection or DR-placement decision recorded here** — the five-geography model, per-geography distribution models, exact-production-region-first placement, `CustomerSeedRecord`, two-region CVAL/DR co-location (PLC-010a) and the Middle East `DR_NOT_OFFERED` position (DEC-001) are all unchanged. The only wording update in this revision is the region-model gate (below), whose legacy “three-region gate” label is replaced with an explicit geography-aware statement so it can no longer be read as a universal three-distinct-regions minimum. The new per-AZ CRG structure (CAP-023) operates **within** a selected region and is specified in ADR-002/ADR-004, not here.

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# ADR-001 - Region Selection and Customer Placement

**Status:** Accepted
**Date:** 27 August 2026
**Deciders:** Principal Cloud Architect, Platform Engineering, DR Owner, Product Platform
**Related requirements:** REG-001..REG-005, PLC-001..PLC-010, RDY-001..RDY-004, DR-014, DR-016..DR-019, DAT-002, DAT-005
**Related constraints:** ENV-003, CAP-011, CAP-012, QUA-012, NFR-002, NFR-010

## Context

Requirements v2.1 changes the placement model from repeated geography-driven selection to a seeded, production-region-first model. Customers and product teams must know the exact Azure production region up front because a broad geography selection creates contract churn, data-residency ambiguity, and inconsistent product placement for the same customer. `[Derived]`

ACRME must still calculate CVAL and DR placement using current capacity, reservation, quota, zone, sharing, restriction, and DR-readiness state. The first valid decision becomes an authoritative seed that future products reuse. `[Decided]`

Key forces:

- Production protection is the primary objective. `[Decided]`
- Region, zone, quota, and capacity are hard isolation boundaries; capacity is never counted across regions or zones. `[Documented]`
- Region strategy is configuration-driven and spans **five supported geographies** (US, EU, Australia, Asia Pacific, Middle East), each catalogued with an explicit **distribution model** (baseline Section 6, REG-001). The US is the only three-region geography; EU, Australia, Asia Pacific and the Middle East are two-region geographies. New regions (e.g. Japan East, pending) are policy data added by config, not automatic rollout commitments. `[Derived]`
- **Middle East DR is currently `DR_NOT_OFFERED` (DR-014, DEC-001 — under legal review).** As it stands, DR is **not offered** in the Middle East: Legal owns the programme and data-sovereignty/residency laws (a largely government/medical customer base) mean cross-border DR cannot meet residency requirements (baseline Section 2, Section 5.2, Section 6). Middle East placement therefore defaults to `dr_region = NOT_OFFERED`; production may still be placed in-geo without DR. Switzerland North is a **pre-configured, conditional** cross-geo extension that activates **only if** Legal clears DEC-001. `[Documented]`
- Placement must return a fresh, machine-readable readiness state to AEP/provisioning and must fail safely on stale or incomplete state. `[Decided]`

# Decision

Adopt a **production-region-first, seeded placement architecture**:

1. **Exact production region is the default input.** The default onboarding path requires the customer or platform workflow to provide the exact Azure production region. ACRME validates it rather than deriving it from a broad geography. `[Decided]`

2. **Geography-only selection is an exception path.** Geography input is retained only with explicit exception approval and customer acknowledgement that the selected production region becomes fixed until an approved migration changes the seed. `[Decided]`

3. **Customer seed record is authoritative.** The first placement decision writes `CustomerSeedRecord`; subsequent products/environments for the same customer and geography reuse it instead of re-running production selection. `[Decided]`

4. **CVAL and DR are selected after production is fixed.** Once the production region is validated, ACRME selects or validates CVAL and DR using current readiness, environment separation, restriction flags, workload distribution, quota/capacity state, and state freshness. `[Derived]`

5. **Placement creates readiness output, not just a region tuple.** Every evaluation returns a `DeploymentReadinessResult` state:

```text
   READY | READY_WITH_RISK | QUOTA_DEFICIT | RESERVATION_DEFICIT |
   CAPACITY_UNAVAILABLE | STALE_STATE | POLICY_BLOCKED | VALIDATION_REQUIRED
```

`[Decided]`

1. **Atomic holds prevent double-commit.** Committed placement creates a hold keyed by customer, region, zone, SKU/family, environment, policy version, and snapshot version. Conditional writes reject concurrent requests that attempt to consume the same headroom. `[Decided]`

2. **DR placement contributes to the DR index.** The seed drives the `SourceDestinationDRIndex` maintained by ADR-003 so a declared source-region failure activates only that source's mapped standby set. `[Derived]`

## Region Classification and Policy

`PlacementPolicy` is the authoritative, versioned region catalogue. It includes region classification, supported geographies, **per-geography distribution model** (three-region vs two-region), zone support, SKU/family eligibility, separation class, approved cross-geo extension paths, `DR_NOT_OFFERED` flags, stale-state thresholds, weights, and exception metadata. `[Decided]`

| Attribute | Requirement |
| --- | --- |
| Standard region | Eligible for automated placement and scoring. |
| Restricted region | Never recommended, scored, or auto-selected; usable only for explicit production-region exception. |
| Cross-geo extension | Usable only when explicitly approved for the source geography. For the **Middle East**, the Switzerland North extension is pre-configured but **inactive** until DEC-001 legal approval clears `DR_NOT_OFFERED`. |
| `DR_NOT_OFFERED` | Produces seed value `DR region = NOT_OFFERED`; ACRME must not silently substitute another geography. **Default `true` for the Middle East** (DR-014, current legal position pending DEC-001) — evaluated **before** any cross-geo extension so an inactive extension is never auto-applied. |
| Policy version | Every change increments version and is recorded with approver, reason, effective date, and replay/audit metadata. |

The number of regions per geography is set by the catalogue's **distribution model**, not a universal minimum. The **US is a three-region geography** — production, CVAL and DR each sit in a distinct region. **EU, Australia, Asia Pacific and the Middle East are two-region geographies**: production is placed in one region and **CVAL and DR are co-located in the other region** (mandatory rule **PLC-010a**, via environment separation ENV-003). A two-region geography therefore delivers in-geo Prod + CVAL + DR **without** any cross-geo path — co-location is the standard model, not a fallback. The **Middle East is the single exception**: DR is `DR_NOT_OFFERED` (DR-014, DEC-001), so its second region hosts CVAL only and production may exist without DR. Cross-geo extension (Switzerland North) applies **only** to the Middle East and only if Legal clears DEC-001. `[Derived]`

## Placement Flow

```
flowchart TD
    Request[Placement request] --> ExistingSeed{Seed exists?}
    ExistingSeed -- Yes --> Reuse[Reuse CustomerSeedRecord]
    ExistingSeed -- No --> Input{Input mode}
    Input -- Exact production region --> ValidateProd[Validate production region]
    Input -- Geography exception --> GeoApproval{Exception approved and acknowledged?}
    GeoApproval -- No --> PolicyBlocked[POLICY_BLOCKED]
    GeoApproval -- Yes --> DeriveProd[Derive production from Standard regions]
    ValidateProd --> ProdOk{Region valid, fresh, supported?}
    DeriveProd --> ProdOk
    ProdOk -- No --> NotReady[Readiness reason]
    ProdOk -- Yes --> SelectCVAL[Select or validate CVAL]
    SelectCVAL --> SelectDR[Select DR or NOT_OFFERED]
    SelectDR --> Fresh{State fresh?}
    Fresh -- No --> Stale[STALE_STATE]
    Fresh -- Yes --> Hold[Create atomic placement hold]
    Hold --> Seed[Write CustomerSeedRecord]
    Reuse --> Readiness[Return DeploymentReadinessResult]
    Seed --> Readiness
```

## CustomerSeedRecord

`CustomerSeedRecord` must include:

- seed ID and customer/realm identifier;
- geography;
- production region;
- CVAL region;
- DR region or `NOT_OFFERED`;
- products covered;
- region/zone/SKU-family policy context;
- decision timestamp;
- policy version and engine version;
- capacity/quota snapshot references and freshness;
- exception approval reference and customer acknowledgement, where applicable;
- active hold IDs;
- migration status and audit metadata. `[Decided]`

Seeds are not regenerated on upgrades, rebuilds, or routine deployments. Changes require an approved migration/exception workflow with impact analysis. `[Decided]`

## Validation Rules

| Rule | Check | Failure state |
|---|---|---|
| Exact Prod default | Request includes a specific production region unless an exception is approved. | `POLICY_BLOCKED` |
| Region catalogue | Region is present in active `PlacementPolicy`. | `VALIDATION_REQUIRED` |
| Standard automated path | Automated selection uses Standard regions only. | `POLICY_BLOCKED` |
| Restricted region exception | Restricted regions require explicit production-only request and approval. | `POLICY_BLOCKED` |
| `DR_NOT_OFFERED` | Geography/country flagged as no-DR writes `NOT_OFFERED` and blocks cross-geo substitution. | `READY_WITH_RISK` or `POLICY_BLOCKED` per policy |
| Region-model gate (geography-aware) | Placement satisfies the geography's **distribution model** (REG-001): a **three-region** geography (US) requires Prod, CVAL and DR in three distinct regions; a **two-region** geography requires Prod in one region and **CVAL + DR co-located** in the other (PLC-010a); a `DR_NOT_OFFERED` geography (Middle East, DEC-001) requires Prod + CVAL only, with no DR region. A universal "three distinct regions" minimum is **not** enforced. | `POLICY_BLOCKED` |
| Freshness | Snapshot age <= configured maximum or synchronous refresh succeeds. | `STALE_STATE` |
| Capacity | Required reserved capacity exists or over-allocation is approved. | `RESERVATION_DEFICIT` or `CAPACITY_UNAVAILABLE` |
| Quota | Consumer/deploying subscription has required quota. | `QUOTA_DEFICIT` |

| Rule | Check | Failure state |
|---|---|---|
| Concurrency | Atomic hold commit succeeds. | `VALIDATION_REQUIRED` |

## Consequences

**Positive**

- Avoids production-region ambiguity and customer contract churn. `[Derived]`
- Produces stable placement across products for the same customer/geography. `[Decided]`
- Makes CVAL/DR decisions replayable from seed, policy version, and snapshot references. `[Decided]`
- Gives AEP/provisioning an explicit readiness contract instead of implicit success/failure. `[Decided]`

**Negative / trade-offs**

- Geography-only onboarding now needs exception governance and customer acknowledgement. `[Decided]`
- Seed migration becomes a governed workflow, not a simple re-run of scoring. `[Derived]`
- `DR_NOT_OFFERED` can create a readiness-with-risk outcome that must be handled commercially and operationally. `[Derived]`
- Atomic holds add state-management complexity but are required to prevent double-committed capacity/quota. `[Decided]`

## Alternatives Considered

| Alternative | Why rejected or constrained |
|---|---|
| Geography selection as default | Caused ambiguous customer intent and inconsistent product placement. `[Derived]` |
| Re-run placement per product | Risks drift across products for the same customer/geography. `[Decided]` |
| Use stale daily/weekly snapshots for deployment | Can deploy into capacity/quota that is no longer available. `[Derived]` |
| Auto-select cross-geo DR for two-region geographies | Rejected. Two-region geographies co-locate CVAL and DR in-geo (PLC-010a); cross-geo substitution is never silent. Where in-geo DR is legally impossible (Middle East), the seed is `DR_NOT_OFFERED`, and any cross-geo path (Switzerland North) is explicit and approval-gated. `[Decided]` |

## Appendix - ADR Summary

| ADR | Requirements Applied | Key Open Items |
|---|---|---|
| ADR-001 Region Selection and Customer Placement | REG-001..005, PLC-001..010a, RDY-001..004, DR-014 | DEC-001 Middle East/no-DR policy; DEC-003 geography exception approver; production stale-state threshold |

## Appendix - Status Legend

| Status | Meaning |
|---|---|
| **Proposed** | Under discussion; not yet ratified |
| **Accepted** | Ratified and in force |
| **Deprecated** | No longer recommended but not yet replaced |
| **Superseded** | Replaced by a later ADR |

## Appendix - Evidence Tag Taxonomy

| Tag | Meaning |
|---|---|
| `[Documented]` | Traceable to Azure platform behaviour or documentation |
| `[Decided]` | An explicit ACRME design choice recorded in this ADR set |
| `[Derived]` | A logical consequence of a documented constraint or decision |
| `[Assumed]` | Architectural judgement pending proof-of-concept validation |

## Related ADRs

- **ADR-002 - Quota and Capacity Management** (`acrme_adr_002_quota_and_capacity_management.md`)
- **ADR-003 - Capacity Management during Disaster Recovery (DR)** (`acrme_adr_003_capacity_management_during_dr.md`)
- **ADR-004 - Forecast, Reconciliation, and Increase of Capacity and Quota** (`acrme_adr_004_forecast_and_increase_of_capacity_and_quota.md`)

**Document Status:** Accepted
**Next Review:** After DEC-001/DEC-003 decisions, readiness API contract review, and first seed-record implementation test.